PREFACE:

We are now on the brink of the fourth industrial revolution—an era that showcases an explosion in the creation and availability of digital information. The requirement for intelligent systems that can access and discover knowledge from large textual sources is higher than ever. As researchers and practitioners in the domain of natural language processing, we find ourselves at the frontier of this technological revolution, where the pursuit of innovative solutions to meet the emerging needs of users and organizations becomes paramount.

Indeed, the idea of the project arose out of the shared enthusiasm for pushing the boundaries of knowledge in the realm of NLP and IR, spurred by a collective vision of endowing LMs with the ability to involve themselves in dynamic interactions with textual data. Taking inspiration from the Retrieval Augmented Generation (RAG) framework, we set out to augment LMs' knowledge base—to allow them to directly retrieve and process external documents during inference. As we move forward with our efforts, we remain committed to the highest quality standards and an unyielding desire for novelty. Our project represents a unique intersection of various methodologies and expertise, making the optimal use of cutting-edge techniques in document loading, document splitting, vector stores and embeddings, and retrieval mechanisms.

In the following sections, we take readers on a thorough journey through all the details of our project, from the basic concepts of document loading and vectorization to the advanced question-answering capabilities of our LM chain.